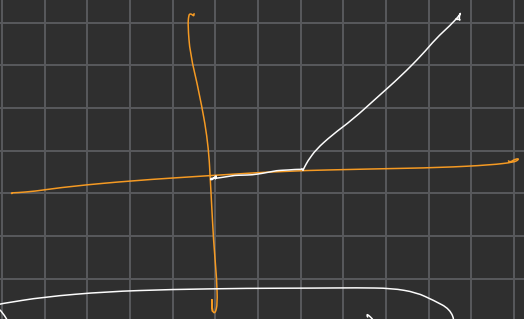
# Activation Function: RELU (Single Perp)

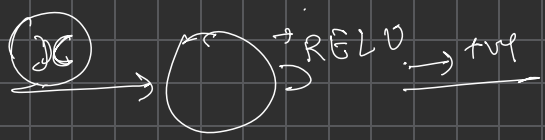
| X | output Y |
|---|---|
| 0 | 0 |
| 1 | 0 |
| 2 | 0 |
| 3 | 0 |
| . | 0 |
| 10 | 0 |
| 11 | 1 |
| 12 | 2 |
| 13 | 3 |



$$\boxed{RELU(x - 10)}$$

output $\leq 0$ → $0$

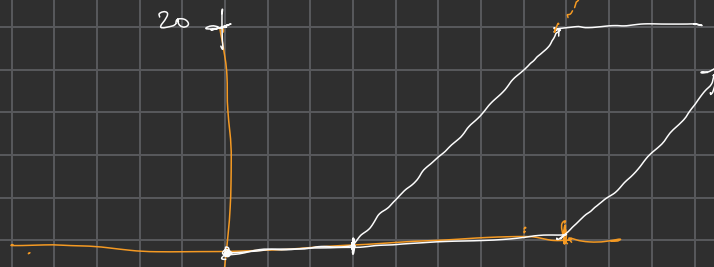
$\geq 0$ → output




$$\text{Output} = w_1 x + b$$

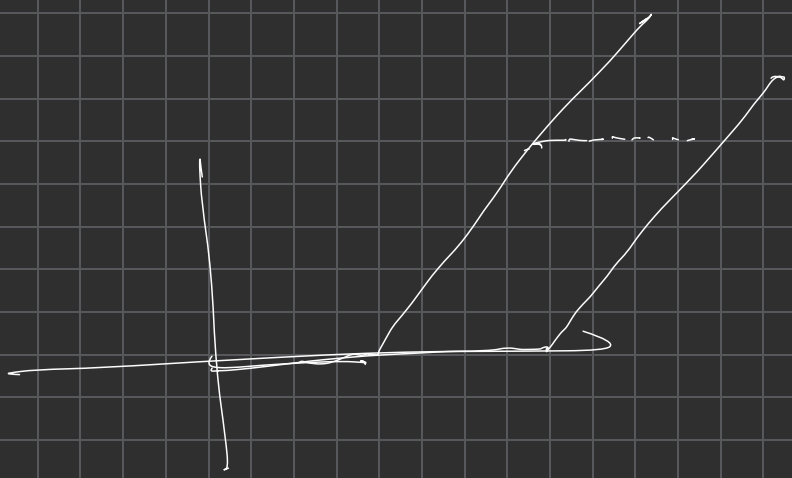
$$= x - 10$$

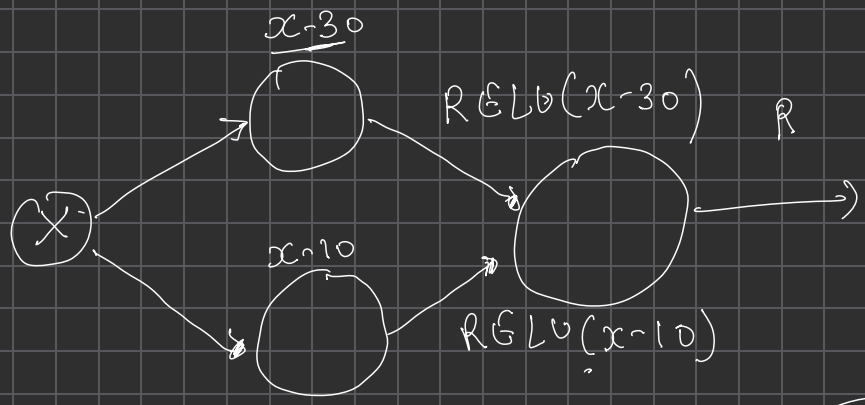
| x | y | | |
|---|---|---|---|
| 0 | 0 | 0 | |
| 1 | 0 | 0 | |
| 2 | 0 | 0 | |
| 3 | 0 | 0 | |
| 4 | 0 | 0 | |
| 10 | 0 | 0 | |
| 11 | 1 | 0 | |
| 12 | 2 | 0 | |
| 13 | 3 | | |
| 14 | 4 | | |
| ⋮ | | | |
| 30 | 20 | 0 | |
| 31 | 20 | 1 | |
| 32 | 20 | 2 | |
| 33 | 20 | 3 | |
| | | 4 | |
| | | 5 | |




RELU(x - 10)

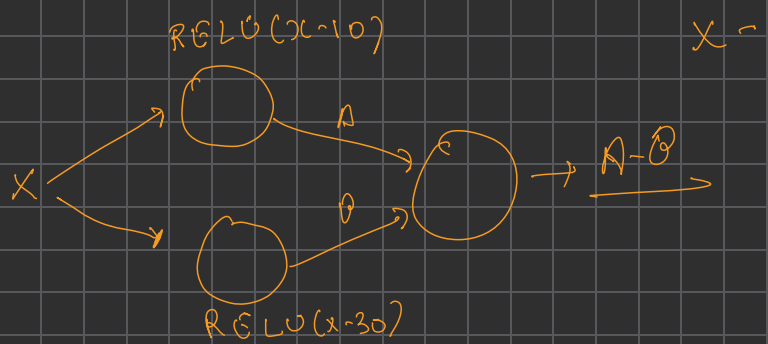
→ RELU(x - 30)





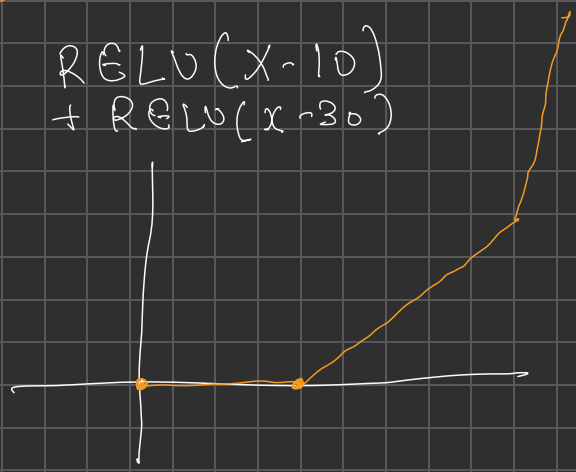

$$w_1 x + w_2 y + B$$

$$\boxed{= \text{RELU}(x-30) + \text{RELU}(x-10)}$$

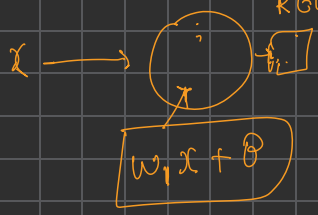


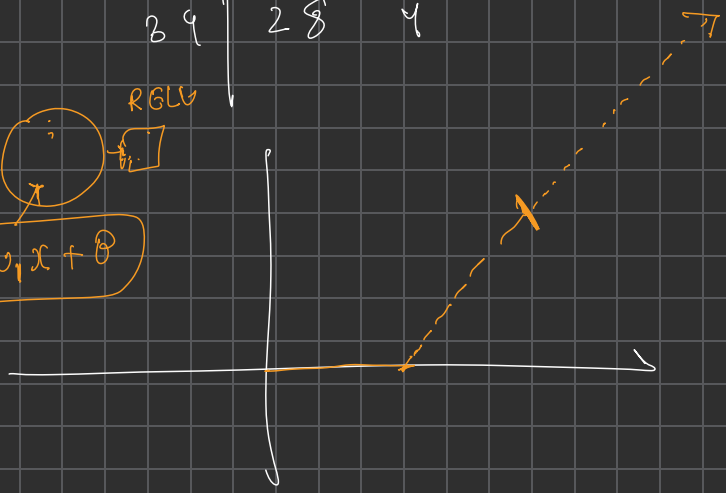
| X | Y | RELU |
|---|---|---|
| 0 | 0 | 0 |
| 1 | 0 | 0 |
| ⋮ | | 0 |
| | | 0 |
| 10 | 0 | |
| 11 | 1 | |
| 12 | 2 | |
| 13 | 3 | |
| ⋮ | | |
| 30 | 20 | 0 |
| 31 | 22 | 1 |
| 32 | 24 | 2 |
| 33 | 26 | 3 |
| 34 | 28 | 4 |

RELU(X-10) + RELU(X-30)



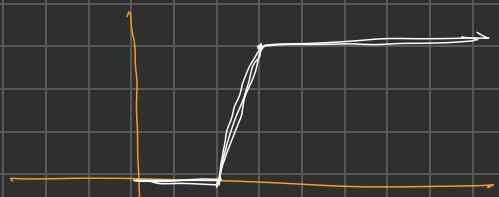
RELU



RELU(x-2)

| X | Y | |
|---|---|---|
| 0 | 0 | 0 |
| 1 | 0 | 0 |
| 2 | 0 | 0 |
| 3 | 3 | 0 |
| 4 | 3 | 3 |
| 5 | 3 | 6 |
| 6 | 3 | 9 |
| 7 | 3 | 12 |
| 8 | 3 | 15 |



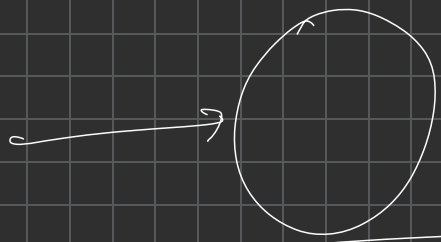
$$\boxed{3*RELU(x-2)} - 3\,RELU(x-3)$$

$$3\,RELU(x-2) - 3\,RELU(x-3)$$

$$RELU(5x-20)$$

| X | Y |
|---|---|
| 0 | 0 |
| 1 | 0 |
| 2 | 0 |
| 3 | 0 |
| 4 | 0 |
| 5 | 5 |
| 6 | 5 |
| 7 | 5 |
| 8 | 5 |

$$5*RELU(x-4) - 5*RELU(x-5)$$

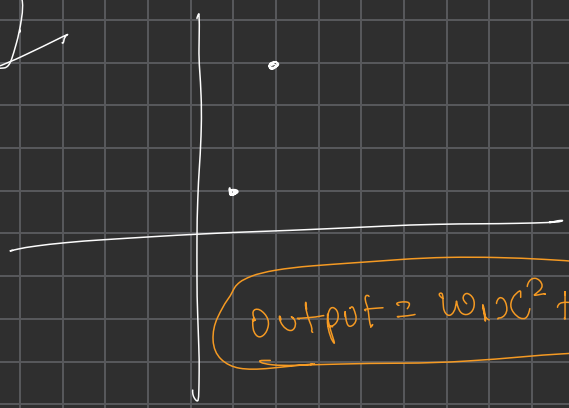


$$\boxed{w_1 x + w_2 y + w_3 z + \theta}$$

$$w_1 x^2 + w_2 x^3 + w_3 x^4 +$$

$$\boxed{Y = x^2}$$

| x | y |
|---|---|
| 1 | 1 |
| 2 | 4 |
| 3 | 9 |
| 4 | 16 |
| 5 | 25 |
| 6 | 36 |
| 7 | 49 |
| 8 | 64 |



$$\boxed{\text{Output} = w_1 x^2 + \theta}$$

$1 \Rightarrow 1$

$2 \Rightarrow 1+3$

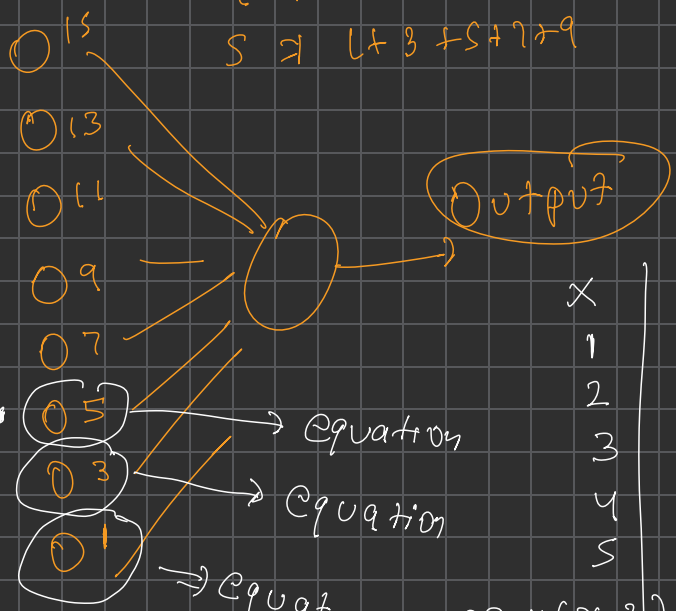
$3 \Rightarrow 1+3+5$

$4 \Rightarrow 1+3+5+7$

$5 \Rightarrow 1+3+5+7+9$

Active = 1

Inactive = 0




x

3

| x | y |
|---|---|
| 1 | 0 |
| 2 | 3 |
| 3 | 3 |
| 4 | 3 |

$3*RELU(x-1) - 3RELU(x-2)$

| x | y output |
|---|---|
| 1 | 0 |
| 2 | 0 |
| 3 | 5 |
| 4 | 5 |
| 5 | 5 |

$5*RELU(x-2) - 5RELU(x-3)$

$$\underline{RELU(x) - RELU(x-1)} + 3(RELU(x-1) - RELU(x-2))$$
$$+ 5(RELU(x-2) - RELU(x-3)) + 7(RELU(x-3) - RELU(x-4))$$
$$+ 9(RELU(x-4) - RELU(x-5)) + \cdots$$

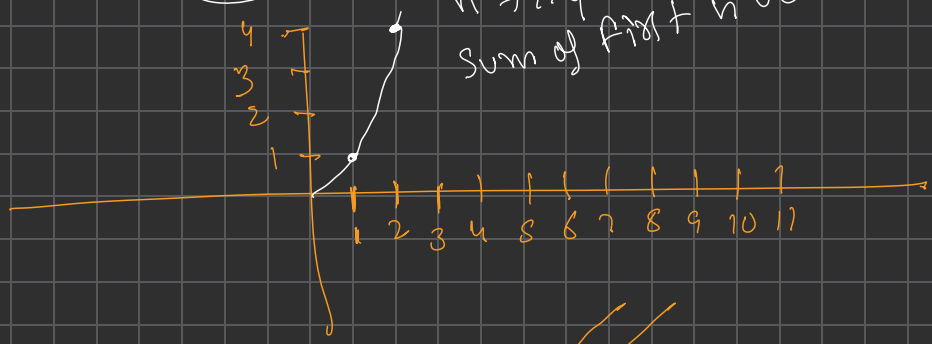
$$RELU(x) + 2\,RELU(x-1) + 2\,RELU(x-2) + 2\,RELU(x-3) + 2\,RELU(x-4)$$

$$[5 + 8 + 6 + 4 + 2]$$ ⇒ Sum of first $(n-1)$ even no + $n$

25

$n \Rightarrow$ sq no

Sum of first $n$ odd numb



6 ⇒

$$6 + 2 + 4 + 6 + 8 + 10$$

36

$$y = x^3$$

| | | |
|---|---|---|
| 1 | 1 | 1 |
| 2 | 8 | 1 + 7 |
| 3 | 27 | 1 + 7 + 19 |
| 4 | 64 | 1 + 7 + 19 + 37 |
| 5 | 125 | 1 + 7 + 19 + 37 + 61 |
| 6 | 216 | 1 + 7 + 19 + 37 + 91 |
| 7 | 343 | 1 + 7 + 19 + 37 + 91 + 127 |
| | | + 169 |

$$y = x^4$$

$$1 * (RELU(x) - RELU(x-1))$$

$$+ 7 RELU(RELU(x-1) - RELU(x-2))$$

$$RELU(x) + 6\,RELU(x-1) + 12\,RELU(x-2) + 18\,RELU(x-3) + 24\,RELU(x-4) + 30\,RE(x-5)$$

$$n^7 \Rightarrow n + \sum_{k=1}^{k=n-1} \left[ 6 * k\ RELU(x-k) \right]$$